AP PRECALCULUS

Unit 1 • Lesson 1.7 practice

Lesson 1.7 • Rational Functions and End Behavior

4 multiple-choice questions and 1 free-response question

Question 1

What is the horizontal asymptote of $r(x) = \dfrac{5x-1}{x^2+4}$?

(A) $y = 5$

(B) $y = 0$

(C) $y = -\dfrac{1}{4}$

(D) There is none.

Question 2

What is the horizontal asymptote of $r(x) = \dfrac{4x^3 - x}{1 - 2x^3}$?

(A) $y = 2$

(B) $y = 4$

(C) $y = -2$

(D) $y = 0$

Question 3

What is the slant asymptote of $r(x) = \dfrac{x^2+3x}{x-1}$?

(A) $y = x + 3$

(B) $y = x + 4$

(C) $y = x$

(D) $y = x - 4$

Question 4

$\lim\limits_{x \to -\infty} \dfrac{x^3}{x^2+1}$ is

(A) 1

(B) 0

(C) $+\infty$

(D) $-\infty$

Question 5 (Free response)

Let $r(x) = \dfrac{6x^2+x}{3x^2-12}$ and $s(x) = \dfrac{2x^2-x+3}{x+1}$.

(a) Write limit statements for the end behavior of r.

(b) Find the equation of the slant asymptote of s.

(c) For large positive values of x, is the graph of s above or below its slant asymptote? Explain.